

React HandsOn 7:-

Theory:

- Define props.

Props (Properties) are a mechanism in React used to pass data from a parent component to a child component. They allow components to communicate with each other and make components reusable by providing different values without changing the component's internal code. Props are read-only (immutable), meaning a child component cannot modify the values it receives from its parent. Instead, it simply uses those values to render the UI. For example, in an online shopping application, a parent component can pass the item name and price to a child component using props, and the child component displays the information without altering it.

- Explain default props.

Default Props are predefined values assigned to a component's props. They are used when a parent component does not pass a value for a particular prop. Default props help prevent errors caused by missing data and ensure that the component still displays meaningful information. They improve the reliability and reusability of components because the component can function correctly even if some props are omitted. For instance, if a product's price is not provided, a default value such as 0 or "Not Available" can be displayed instead of leaving the field blank or causing an error.

- Define the difference between state and props.

Features	Props	States
Definition	Props (Properties) are used to pass data from a parent component to a child component.	State is used to store and manage data within a component.
Ownership	Owned and controlled by the parent component.	Owned and controlled by the component itself.
Mutability	Props are read-only (immutable) and cannot be modified by the child component.	State is mutable and can be updated using <code>setState()</code> (class components) or <code>useState()</code> (functional components).

Purpose	Used for communication and sharing data between components.	Used to manage dynamic data such as user input, button clicks, form values, or API responses.
Data Flow	Data flows from parent to child (one-way data binding).	Data remains within the component unless passed to other components through props.
Re-rendering	A component re-renders when the parent passes new props.	A component re-renders whenever its state changes.
Example	Passing a product name and price from a parent component to a child component.	Updating a counter value when a button is clicked or storing the user's input in a form.

- Explain ReactDOM.render().
ReactDOM.render() is a method used to render a React component into the Document Object Model (DOM) of a web page. It connects the React application to the HTML element, usually the `<div id="root"></div>` present in the `public/index.html` file. This method tells React which component should be displayed first in the browser. In React 17 and earlier versions, applications commonly used `ReactDOM.render(<App />, document.getElementById('root'))`; Starting from React 18, this method has been replaced by `createRoot()` for better performance and improved support for concurrent rendering. The new syntax is:

Practical:

- Code:

```

JS TrainersMock.js U  JS OnlineShopping.js U  JS App.js M X  JS index.js M  # index.css M  JS Cart.js U
1  import OnlineShopping from "../OnlineShopping";
2  function App()
3  {
4      return (
5          <div>
6              <OnlineShopping />
7          </div>
8      );
9  }
10 export default App;
11

```

```
TrainersMock.js U JS OnlineShopping.js U JS App.js M # App.css X JS index.js M # index.css M JS Cart.js
1 .App {
2   text-align: center;
3 }
4
5 .App-logo {
6   height: 40vmin;
7   pointer-events: none;
8 }
9
10 @media (prefers-reduced-motion: no-preference) {
11   .App-logo {
12     animation: App-logo-spin infinite 20s linear;
13   }
14 }
15
16 .App-header {
17   background-color: #282c34;
18   min-height: 100vh;
19   display: flex;
20   flex-direction: column;
21   align-items: center;
22   justify-content: center;
23   font-size: calc(10px + 2vmin);
24   color: white;
25 }
26
27 .App-link {
28   color: #61dafb;
29 }
30
31 @keyframes App-logo-spin {
32   from {
33     transform: rotate(0deg);
34   }
35   to {
36     transform: rotate(360deg);
37   }
38 }
```

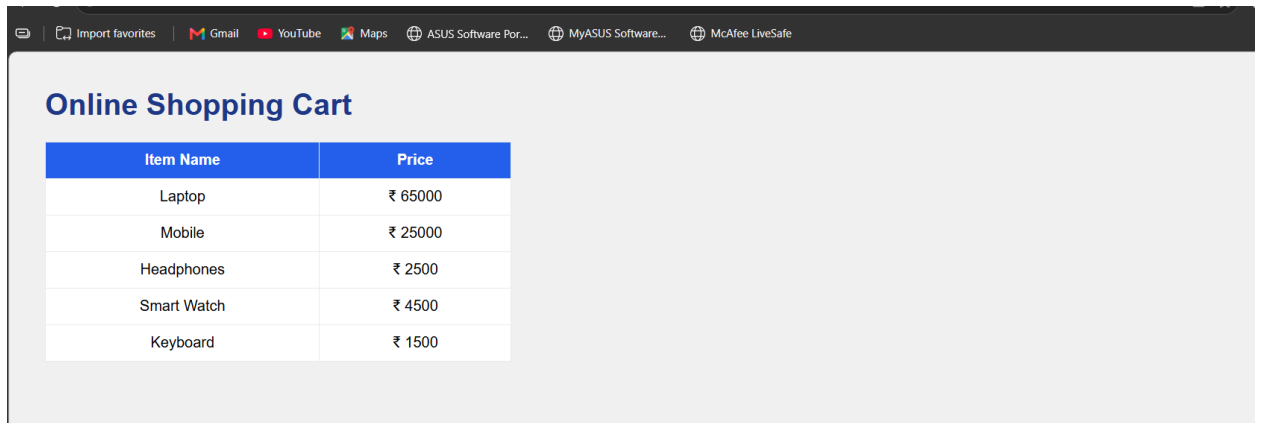
```
JS TrainersMock.js U JS OnlineShopping.js U JS App.js M # App.css JS index.js M # index.css M
1 import React, { Component } from "react";
2
3 class Cart extends Component {
4   render() {
5     return (
6       <tr>
7         <td>{this.props.Itemname}</td>
8         <td>₹ {this.props.Price}</td>
9       </tr>
10     );
11   }
12 }
13 export default Cart;
```

```
JS TrainersMock.js U JS OnlineShopping.js U JS App.js M # App.css JS index.js M X # index.css M
1 import React from "react";
2 import ReactDOM from "react-dom/client";
3 import "./index.css";
4 import App from "./App";
5
6 const root = ReactDOM.createRoot(document.getElementById("root"));
7 root.render(
8   <React.StrictMode>
9     <App />
10   </React.StrictMode>
11 );
12
```

```
JS TrainersMockjs U JS OnlineShopping.js U X JS App.js M # App.css JS index.js M # index.css M JS Cart.js U
1 import React, { Component } from "react";
2 import Cart from "../Cart";
3
4 class OnlineShopping extends Component {
5   constructor() {
6     super();
7     this.items = [
8       {
9         Itemname: "Laptop",
10        Price: 65000
11      },
12      {
13        Itemname: "Mobile",
14        Price: 25000
15      },
16      {
17        Itemname: "Headphones",
18        Price: 2500
19      },
20      {
21        Itemname: "Smart Watch",
22        Price: 4500
23      },
24      {
25        Itemname: "Keyboard",
26        Price: 1500
27      }
28    ];
29  }
}
```

```
31 render() {
32   return (
33     <div>
34       <h1>Online Shopping Cart</h1>
35       <table border="1" cellPadding="10">
36         <thead>
37           <tr>
38             <th>Item Name</th>
39             <th>Price</th>
40           </tr>
41         </thead>
42         <tbody>
43           {this.items.map((item, index) => (
44             <Cart
45               key={index}
46               Itemname={item.Itemname}
47               Price={item.Price}
48             </>
49           ))}
50         </tbody>
51       </table>
52     </div>
53   );
54 }
55
56 export default OnlineShopping;
```

- Output:



Item Name	Price
Laptop	₹ 65000
Mobile	₹ 25000
Headphones	₹ 2500
Smart Watch	₹ 4500
Keyboard	₹ 1500